



(High Resolution)

Title: (Genesis of Lithium-rich Brines in the Jiangnan Basin: A "Multi-source and Multi-driver" Enrichment Model)

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Abstract: Deep-seated brines in the Jiangnan Basin represent a strategic lithium resource with significant industrial potential. Analytical results reveal Li concentrations of 22–261 mg/L, a high Li/Mg ratio (10–20), and co-enrichment of K (1–4 g/L), Br (100–900 mg/L), and B (40–930 mg/L). This study elucidates the synergetic "multi-source and multi-driver" enrichment mechanism through geochemical signatures, mineralogical evaluation, and multi-isotopic tracers (H, O, S, B, Li).

Multi-isotopic evidence provides robust constraints on fluid evolution. H-O isotopes (δD avg. -61.2‰; $\delta^{18}O$ avg. -5.73‰) reflect the mixing of meteoric water and concentrated liquors along an evaporation trend. Enriched $\delta^{34}S$ (+11.4 to 53.5‰) indicates intense thermochemical sulfate reduction (TSR) of sedimentary gypsum. Furthermore, low $\delta^{11}B$ (+9 to 41‰) and δ^7L (+9 to 22‰) values, significantly below seawater averages, correlate with continental geothermal fluids and deep-seated hydrothermal inflows, confirming a multi-source origin.

The "multi-driver" mechanism is centered on temperature-controlled mineralogical transformations. Mineralogical data show that Li concentration is positively correlated with chlorite, illite, and smectite-suggesting initial adsorption-but strongly negatively correlated with illite/smectite (I/S) mixed-layer minerals. At the critical burial depth of 2500–3000 meters, the rising geothermal gradient triggers the transformation of smectite to illite, releasing structurally adsorbed lithium into the brines. Beyond 3000 meters, Li grades continue to increase as water-rock interactions intensify, further augmented by hydrothermal heat in regions like the Jiangling Sag.

In conclusion, we propose a "Source-Heat-Dynamics" model where surface pre-concentration is enhanced by deep-burial processes. The I/S mixed-layer zone and low $\delta^{77}L$ signatures are established as key mineralogical and geochemical markers for Li-rich brine

exploration. This framework offers new theoretical guidance for assessing deep-seated brine resources in continental salt lake basins.

Keywords: Jiangnan Basin; Lithium-rich brine; Multi-isotope tracers; Clay mineral transformation; I/S mixed-layer; 2500–3000m threshold

Biography:

Dr. Zhen Yang obtained his Bachelor's degree in Materials Science and Engineering from China University of Geosciences (Wuhan) and his PhD degree in Mineral Deposits from the same university. He has worked as an exploration geologist focusing on the metallogenic mechanisms and exploration of lithium-rich brine deposits in the Jiangnan Basin and Jitai Basin. His research centers on the application of multivariate geochemistry and integrated information methods to study the mineralization processes and exploration of deep lithium-rich brines in sedimentary basins. Currently, he lectures on Mineral Deposits and Mineral Prospecting at China University of Geosciences (Wuhan). His research interests include brine mineral deposits in basins, hydrothermal deposits, and brine development. He has published multiple papers in peer-reviewed journals.

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